

# Ultracompact and large-bandwidth silicon modulator in a CMOS-compatible foundry

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The exponential growth of global data traffic demands transformative solutions for data centers. While optical interconnects offer a promising pathway to overcome the bandwidth limitation, conventional electro-optic modulators face a bandwidth-footprint trade-off. Here, we demonstrate a silicon modulator by leveraging the slow-light effect in a photonic crystal nanobeam cavity. The proposed modulator exhibits a 110-GHz electro-optic bandwidth with an ultracompact footprint of  $10\ \mu\text{m}^2$  while enabling precise light manipulation with an 80 pm/V tuning efficiency within a  $0.12\ \mu\text{m}^3$  mode volume. We also demonstrate 110-Gbps and 130-Gbps non-return-to-zero signal transmissions with bit error rates less than  $3.8 \times 10^{-3}$  and  $2 \times 10^{-2}$ , respectively, while maintaining a power consumption of 5.9 fJ/bit. This breakthrough represents an important advance in photonic interconnect technology, as it overcomes the critical trade-off between bandwidth and footprint while achieving an ultra-high energy efficiency, thereby paving the way for next-generation ultracompact, high-speed, and energy-efficient optical interconnects in data centers and beyond.

The continuous expansion of cloud computing and artificial intelligence infrastructures<sup>1</sup> has precipitated an unprecedented increase in global data traffic, and the fundamental limitations of copper-based electrical interconnects in modern hyperscale data centers and exascale computing architectures have been exposed. This technological impasse has intensified the search for photonic alternatives capable of sustaining terabit-level throughput with energy-efficient operation. Among various photonic integration platforms, silicon photonics stands out as a promising candidate<sup>2</sup>, with the merits of a compact footprint and compatibility with mature complementary-metal-oxide-semiconductor (CMOS) process technology. These attributes enable wafer-scale integration of photonic-electronic systems, making the silicon platform a cornerstone for next-generation optical networks.

Central to the silicon photonic revolution is the development of high-performance electro-optic (EO) modulators<sup>3–7</sup>, which have been extensively investigated by leveraging the plasma dispersion effect. Mach–Zehnder interferometer (MZI) modulators (MZMs) typically

exhibit a large footprint, with phase shifter arms extending to the millimeter scale to accumulate sufficient phase differences<sup>8–13</sup>. For example, in 2021, a silicon MZM with a footprint of  $0.9\ \text{mm} \times 2.2\ \text{mm}$  was presented, which allowed 220 Gbps/λ net rate operation<sup>14</sup>. The large footprints of MZMs pose challenges for achieving high-density integration and low-power consumption. To address these issues, various approaches have been explored, including the use of organic-silicon hybrid platforms<sup>15–17</sup>, slow-wave enhancement<sup>18,19</sup>, and heterogeneous integration with other materials<sup>20–22</sup>. These strategies aim to enhance the interactions between the optics and materials, thereby shrinking the footprint and improving the tuning efficiency. Recently, Han et al.<sup>23</sup> demonstrated a high-speed slow-light silicon MZM with a large EO bandwidth of 110 GHz, in which the phase shifter length was reduced to only 124 μm. Nevertheless, owing to the constraint imposed by the width of the traveling-wave electrodes, further reduction of the footprint remains infeasible. Alternatively, microring resonator (MRR)-based modulators<sup>24–27</sup> (MRMs) have advantages in terms of compactness and energy efficiency. Xu et al.<sup>28</sup> demonstrated the first silicon

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MRM driven by a p-i-n junction. However, the operation speed was limited by the long free-carrier lifetime (~1 ns) within the diodes<sup>2</sup>. In 2022, a carrier depletion-based resonator modulator<sup>29</sup> was proposed, achieving an EO bandwidth larger than 67 GHz. Chan et al.<sup>30</sup> demonstrated efficient 330-Gbps modulation using a silicon MRM with a 50-GHz bandwidth. Despite extensive efforts, the requirements for integrated optical modulators are becoming increasingly stringent as the demands for high capacities and compact footprints generally continue to grow<sup>31</sup>. The simultaneous realization of an ultracompact footprint and a large bandwidth in a silicon modulator while maintaining compatibility with standard CMOS manufacturing processes remains a significant challenge. Addressing this challenge requires innovative approaches that leverage advanced photonic structures to enhance light–matter interactions within a compact device footprint and theoretical analysis to broaden the operation bandwidth.

A one-dimensional (1D) photonic crystal nanobeam cavity (PCNC) has emerged as a promising solution, offering an ultracompact footprint and a wavelength-scale mode volume<sup>32–35</sup>. Unlike MZIs and MRRs, which exhibit a constant optical distribution in their waveguides, the PCNC features a high electromagnetic energy density and a high optical concentration factor<sup>36</sup>. These unique properties significantly enhance the interactions between light and materials, enabling precise control of the optical field within a smaller active material volume. In fact, the effective mode volume in a 1D PCNC is orders of magnitude lower than that in MRRs<sup>37</sup>, making it an ideal candidate for compact and efficient modulators. However, previous silicon modulators utilizing PCNCs achieved bandwidths and modulation rates below 40 GHz and 70 Gbps<sup>38–40</sup>, respectively, highlighting the need for further theoretical exploration of the interplay between the Q factor, tuning efficiency, and EO bandwidth, which is critical for advancing future high-speed optical interconnect technologies.

In this work, we investigate the slow-light enhancement in an optical resonator and present a comprehensive theoretical framework for designing a resonance-based EO modulator with a large bandwidth. Notably, the analytical method proposed here is not limited to silicon modulators but is also applicable to other materials, including LiNbO<sub>3</sub> (LN)<sup>41</sup>, BaTiO<sub>3</sub> (BTO)<sup>42</sup>, and EO polymers<sup>43</sup>. Building on the slow-light effect and theoretical analysis, we design and fabricate a PCNC-based EO modulator in a standard silicon photonic commercial CMOS-compatible foundry. With the slow-light enhancement in the resonant cavity, the inherently weak light–matter interactions are significantly amplified, yielding a 3.36-fold improvement in phase detuning compared with conventional silicon rib waveguide cavities in our simulation, which were designed with identical waveguide width, length, and slab thickness. Importantly, the slow-light effect amplifies the phase shift per unit voltage rather than directly altering the material's electro-optic properties. This enhancement, combined with an appropriate doping concentration provided by the commercial foundry, enables a tuning efficiency of 80 pm/V for our optimized PCNC. Through precise control of the resonator's dynamic response via carrier-wavelength tuning, we exploit the optical peaking enhancement<sup>44</sup> to optimize the trade-off between photon-lifetime and resonance conditions<sup>44–47</sup>. This approach effectively extends the EO bandwidth from 59 GHz to 110 GHz with only a 1.84 dB roll-off within an ultracompact footprint of 10 μm<sup>2</sup>. To the best of our knowledge, the demonstrated modulator has an EO bandwidth exceeding 110 GHz with the most compact device footprint on the silicon platform. As an application example, the proposed modulator is operated with a non-return-to-zero (NRZ) signal of up to 110 GBaud, with a bit error rate (BER) less than  $3.8 \times 10^{-3}$ , setting a benchmark for compact, high-performance photonic devices.

## Results

### Device principle and design

The core objective of this work is to realize a compact and large-bandwidth modulator on the silicon photonic platform to address the

critical demands for future high-density integration and high-speed single-lane data transmission. To achieve this, we propose a 1D PCNC with a tailored grating unit cell, leveraging the slow-light effect to enhance light–matter interactions within the resonant cavity. Below, we systematically elucidate the design principles and performance trade-offs underpinning this device.

Universally, the overall EO bandwidth ( $BW_{EO}$ ) of a resonance-based modulator is governed by the interplay between the optical and electrical bandwidths, as defined by<sup>48</sup>

$$\frac{1}{BW_{EO}^2} = \frac{1}{BW_O^2} + \frac{1}{BW_E^2} \quad (1)$$

where the optical bandwidth ( $BW_O$ ) is inversely proportional to the Q factor of the cavity, and the electrical bandwidth ( $BW_E$ ) depends on the resistance–capacitance (RC) time constant of the driving circuit (Supplementary Information Note 1). To realize a large EO bandwidth,  $BW_E$  and  $BW_O$  should be simultaneously optimized. While a low Q factor inherently broadens the  $BW_O$ , it simultaneously necessitates greater phase detuning to achieve a defined dynamic extinction ratio (ER). However, in high-speed EO modulation on the silicon platform, it is typically implemented using a reverse-biased PN junction, whose tuning efficiency is generally constrained by the specifications of a standard CMOS-compatible foundry process. Consequently, a low Q factor in the resonator leads to a reduction in the dynamic ER. This reduction can significantly lower the signal quality and increase the BER, which is a critical consideration for practical applications. To mitigate the adverse effects of a low Q factor on the dynamic ER, an effective strategy is to improve the tuning efficiency of the modulator, thereby optimizing its performance under these constraints.

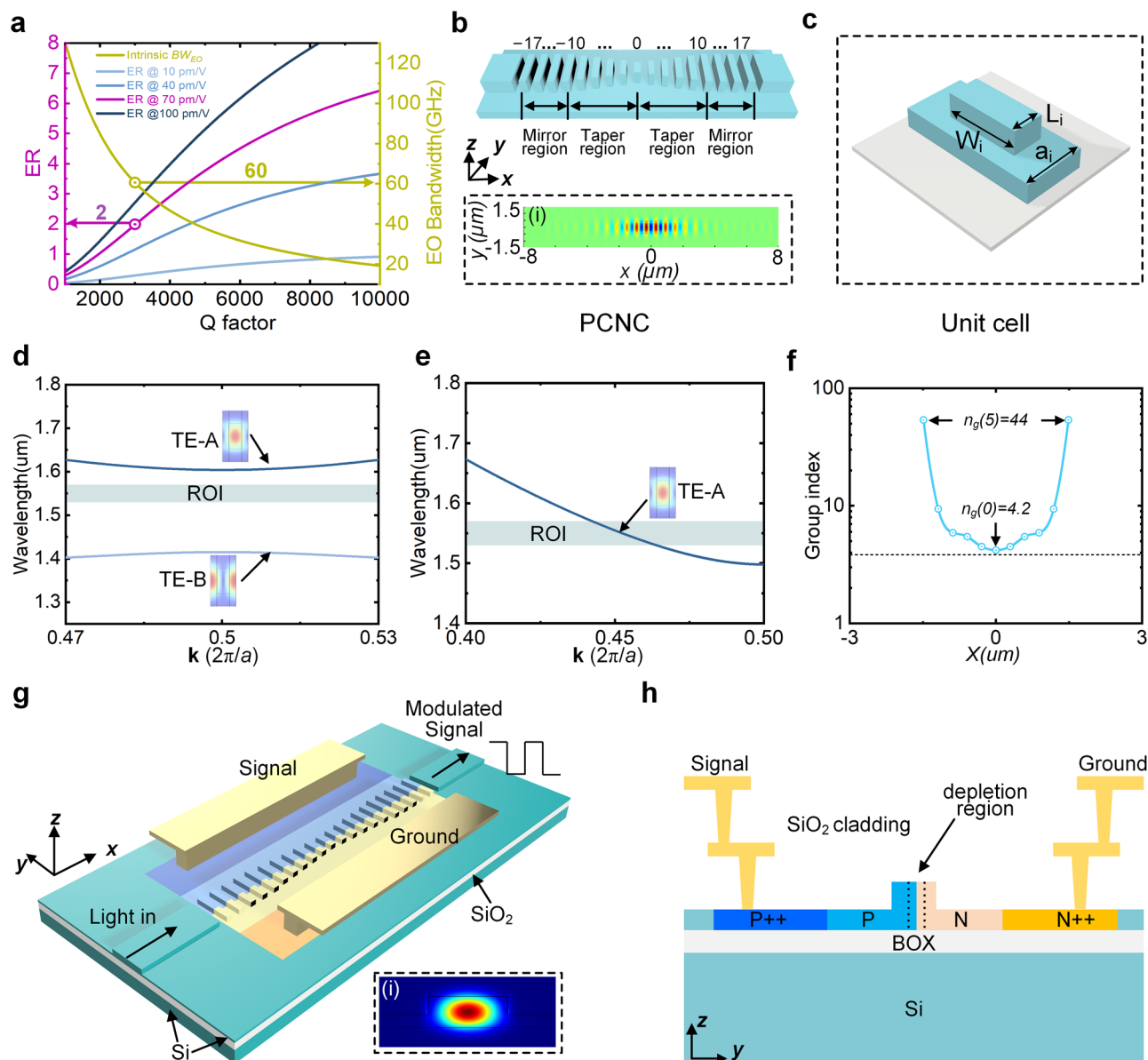
In Supplementary Information Note 2, we systematically analyze the interplay between the dynamic ER, EO bandwidth, Q factor, and tuning efficiency. Through rigorous optimization, we identify a balanced trade-off by setting the Q factor to 3000, corresponding to a photon-lifetime of  $\tau = Q/\omega = 2.5$  ps. This approach limits the optical bandwidth to 63.7 GHz, yielding an EO bandwidth of 60 GHz (obtained from Eq. 1). To extend the modulation bandwidth, the optical peaking enhancement can be utilized in the modulator as reported in previous works<sup>5,41</sup>, which is achieved by strategically detuning the operating wavelength from the cavity's resonance peak. Additionally, as shown in Fig. 1a, the dynamic ER is directly affected by the tuning efficiency with a given swing voltage (Supplementary Information Note 2). To achieve a dynamic ER greater than 2 dB while maintaining the intrinsic EO bandwidth of 60 GHz, the tuning efficiency must be increased to approximately 70 pm/V, which is several times higher than that in a previous work<sup>40</sup>.

To significantly improve the tuning efficiency of the modulator, particularly in devices leveraging the plasma dispersion effect, the slow-light effect<sup>49–51</sup> has been widely adopted. The accumulated phase detuning in the slow-light waveguide is proportional to the group index  $n_g$ , as described by<sup>23</sup>

$$\phi = \eta \cdot n_g \cdot L \cdot \frac{2\pi}{\lambda_0} \quad (2)$$

where  $\eta = \eta_0 (\sqrt{V + V_{bi}} - \sqrt{V_{bi}})$ , with  $\eta_0$  being a coefficient related to the overlap between the carriers and the optical field;  $V_{bi}$  is the built-in voltage of the PN junction;  $V$  is the applied voltage of the modulator; and  $L$  is the optical travel length. This relationship underscores the importance of the slow-light effect in improving the tuning efficiency.

Hence, we introduce the slow-light effect into a compact PCNC, which is designed on a 220-nm silicon-on-insulator (SOI) platform with a 90-nm slab layer to facilitate doping and electrical contact, as shown in Fig. 1b. The PCNC consists of two mirror regions and two taper regions. The mirror regions are carefully designed with a period  $a$  to



**Fig. 1 | Design of the modulator on the silicon platform.** **a** Analytical results of the fundamental functions among the Q factor, dynamic extinction ratio, and electro-optic bandwidth with various tuning efficiencies. The data marked by the circle correspond to  $Q=3000$ . **b** Schematic diagram of the photonic crystal nanobeam cavity; the inset shows the simulated optical field distribution of the photonic crystal nanobeam cavity at resonance. **c** Basic unit cell of the one-dimensional

photonic crystal nanobeam cavity. Band diagram of **d** the mirror regions and **e** the center unit cell in the taper regions. **f** Group index  $n_g \geq 4.2$  within the cavity along the  $x$ -axis. The dashed line indicates the group index of a conventional silicon rib waveguide for comparison. **g** Three-dimensional schematic diagram of the designed modulator; the inset depicts the mode field in the rib waveguide. **h** Cross section of the modulator.

create a photonic bandgap in the region of interest (ROI). The unit cell of the PCNC is shown in Fig. 1c. The band diagram of the mirror regions is obtained by employing a finite element method. A photonic bandgap of 190 nm is generated at the boundary of the Brillouin zone, as illustrated in Fig. 1d. To obtain a resonance of transverse electric mode A (TE-A), the period of the taper regions linearly decreases from  $a$  to  $0.93a$  over 10 periods (details are provided in Supplementary Information Note 3), shifting the energy band to shorter wavelengths. The band diagram of the center unit cell in the taper regions is depicted in Fig. 1e, covering the ROI with the TE-A energy band. As shown in Fig. 1f, light is progressively slowed within the resonant cavity with a group index of  $4.2 \leq n_g \leq 44$ . For reference, the dashed line indicates  $n_g = 3.9$  for a conventional silicon rib waveguide cavity with identical waveguide width, length, and slab thickness design. The group index is calculated from the band diagram of the  $i$ -th cell using the well-known

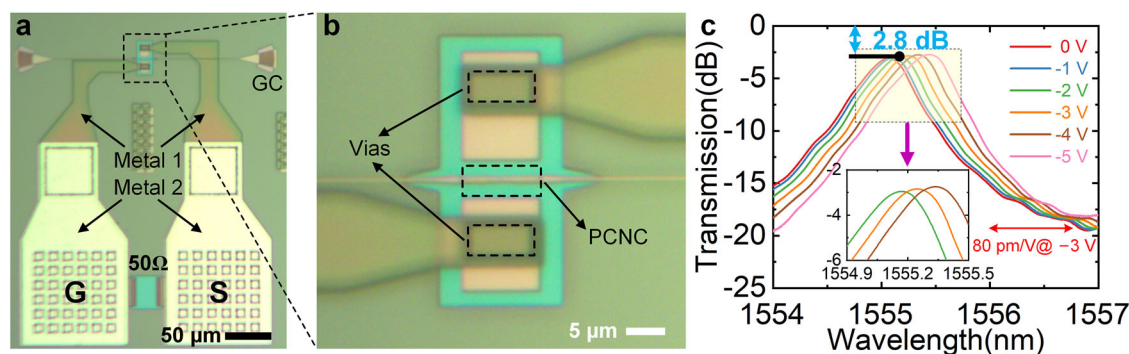
formula

$$n_g = \left| c \frac{dk}{d\omega} \right| = \left| \frac{\lambda^2}{2\pi} \frac{dk}{d\lambda} \right| \quad (3)$$

where  $\lambda$  is the wavelength and  $\mathbf{k}$  is the wavevector. In the  $i$ -th cell of the resonant cavity, the phase accumulation is enhanced by the slow-light effect by a factor of  $n_g(i)$ . Thus, in such a designed PCNC, the total phase accumulation can be derived as follows:

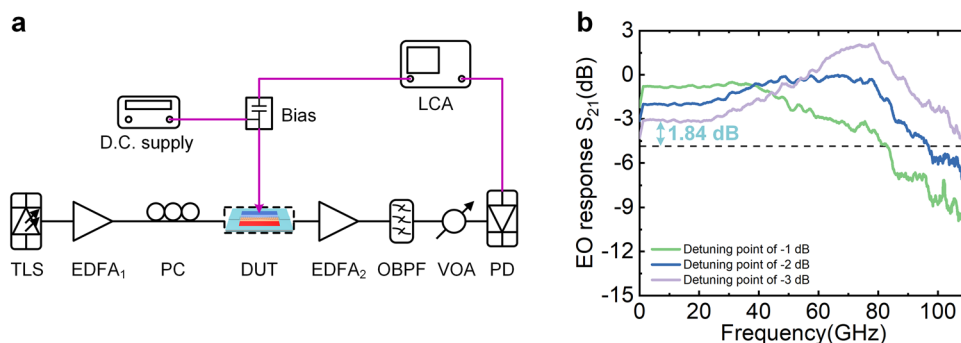
$$\phi = \eta \cdot \frac{2\pi}{\lambda_0} \cdot \sum_{i=-s}^s n_g(i) \cdot a(i) \quad (4)$$

where  $n_g(i)$  and  $a(i)$  are the group index and period of the  $i$ -th cell of the PCNC, respectively. Notably, the summation range is determined



**Fig. 2 | Fabrication and static characterization of the proposed modulator.** **a** Optical microscope image of the fabricated photonic crystal nanobeam cavity modulator. **b** Magnified optical image of the center of the proposed device.

**c** Measured transmission spectra versus reverse bias voltage ranging from  $-1$  V to  $-5$  V; the inset shows the detailed optical spectra under the bias of  $-2$  V,  $-3$  V, and  $-4$  V, respectively. G ground, S signal.



**Fig. 3 | Electro-optic bandwidths of the proposed modulator.** **a** Experimental setup for the electro-optic bandwidth measurement. **b** Measured electro-optic  $S_{21}$  responses for different wavelength detuning  $\Delta\lambda$  values. TLS tunable laser source,

EDFA erbium-doped fiber amplifier, PC polarization controller, DUT device under test, VOA variable optical attenuator, OBPF optical bandpass filter, PD photo-detector; and LCA lightwave component analyzer.

by the specific energy bands and bandgaps of the device (detailed in Supplementary Information Note 7). In contrast to conventional Fabry–Pérot (FP) cavity modulators that depend exclusively on mirror reflectivity for optical enhancement, our design incorporates an engineered slow-light region to combine both enhancement mechanisms: the FP resonance boosts light–matter interaction through recirculating passes, and the slow-light structure offers enhancement by extending the effective interaction length<sup>50,52</sup>, leading to increased tuning efficiency. In addition, the optical field is tightly confined within a mode volume of  $0.12 \mu\text{m}^3$ , enabling precise control over the light in a compact active material region.

A 3D schematic diagram of the proposed modulator is depicted in Fig. 1g. A PN junction is laterally integrated onto the silicon rib waveguide, matching the length of the PCNC to improve the tuning efficiency. The cross section of the modulator, depicted in Fig. 1h, features a slightly wider p-doped region relative to the n-doped region, which increases the overlap between the depletion region and the optical field. Heavily doped regions (P++ and N++) are used to achieve ohmic contact between the metal vias and the silicon layer. The resistance and capacitance of the PN junction are designed to be  $150 \Omega$  and  $5.9 \text{ fF}$ , respectively, resulting in an RC-limited bandwidth of  $180 \text{ GHz}$  (Supplementary Information Note 2).

### Fabrication and static characterization

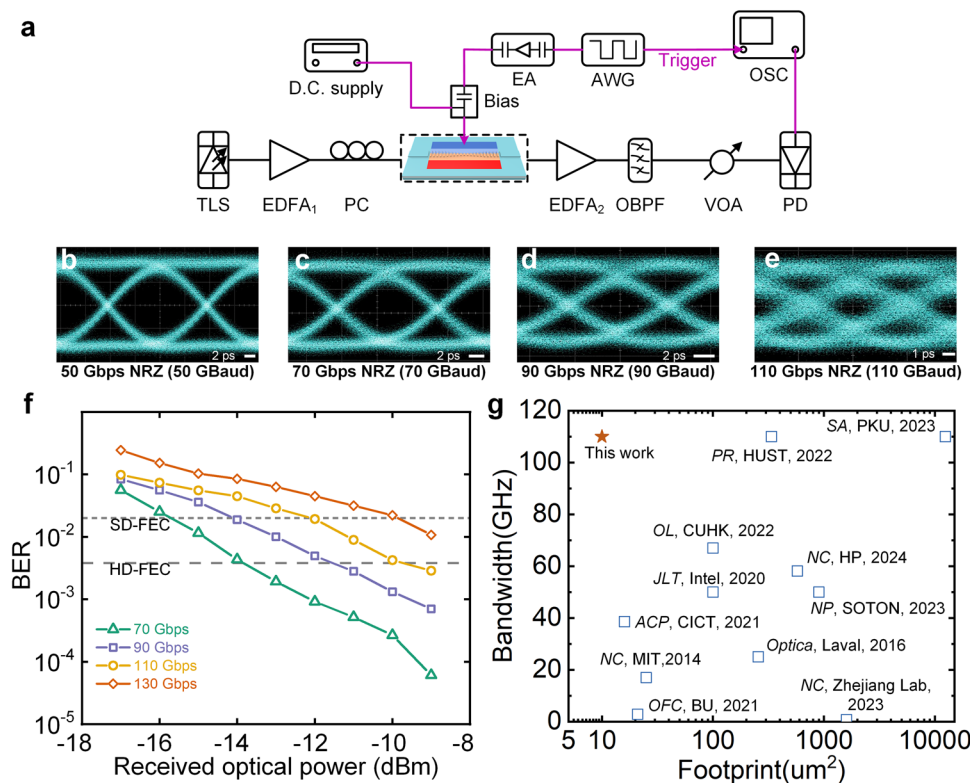
The device was fabricated on a  $220\text{-nm}$  SOI platform in a standard CMOS-compatible foundry at Advanced Micro Foundry (AMF) in Singapore (see “Methods” for details of device fabrication). Figure 2a presents an optical microscope image of the fabricated modulator. To ensure efficient impedance matching and light coupling, a  $50\text{-}\Omega$

resistor and grating couplers (GCs) from the AMF process design kit (PDK) were integrated into the design. A detailed view of the device is shown in Fig. 2b. The static EO response of the modulator was characterized by measuring the transmission spectra under reverse bias voltages ranging from  $-1$  V to  $-5$  V, as illustrated in Fig. 2c. A DC probe was employed to apply the voltage to the bond pad. Under a reverse bias, the electron and hole concentrations in the PN junction decrease, leading to a change in the effective refractive index of the silicon waveguide. This results in a redshift of the resonance wavelength of the PCNC due to the effective increase in the cavity length. Key performance metrics were measured, including a  $3\text{-dB}$  bandwidth of  $0.53 \text{ nm}$ , an insertion loss (IL) of  $2.8 \text{ dB}$ , and a Q factor of  $2930$ . As the reverse bias voltage increases, the depletion region of the PN junction widens, further reducing the carrier concentration and slightly decreasing the IL of the PCNC. At a reverse bias voltage of  $-3$  V, the tuning efficiency was approximately  $80 \text{ pm/V}$ . This value was obtained from a  $160 \text{ pm}$  spectral redshift measured from  $-2$  V to  $-4$  V, corresponding to a  $V_{\pi} \cdot L$  product of  $-0.38 \text{ V}\cdot\text{cm}^2$ . Compared with previous work<sup>40</sup>, the tuning efficiency is improved by 8 times with the slow-light enhancement in our proposed device. These results demonstrate the ability of the device to achieve efficient wavelength tuning with low driving voltages, highlighting its potential for high-performance modulation in integrated photonic circuits.

### High-speed operation characterization

To assess the high-speed modulation performance of the proposed device, an experimental setup to measure the small-signal response ( $S_{21}$ ) was designed, as illustrated in Fig. 3a. A tunable continuous-wave laser (Santec TSL-710) served as the light source, with its optical power





**Fig. 4 | High-speed data modulation measurement and performance comparison.** **a** Experimental equipment for measuring optical eye diagrams. Eye diagrams for non-return-to-zero signals at data rates of **b** 50 Gbps, **c** 70 Gbps, **d** 90 Gbps and

**e** 110 Gbps. **f** Measured curves of the bit error rate versus received optical power for the non-return-to-zero signals at data rates of 70, 90, 110, and 130 Gbps. **g** Review of silicon modulators: electro-optic bandwidths and footprints.

amplified by an erbium-doped fiber amplifier (EDFA). Another EDFA was used to amplify the optical signal modulated by the PCNC modulator to compensate for the optical IL. After that, an optical bandpass filter (OBPF) was employed to reduce the amplified spontaneous emission noise. Prior to detection by a 90 GHz photodetector (PD, XPDV4120R-WFFP), a variable optical attenuator (VOA) was utilized to adjust the received optical power. The output signal from the PD was analyzed using a 110 GHz bandwidth lightwave component analyzer (LCA, Keysight N5227B). A high-speed electrical signal with a sweeping frequency from the LCA was applied to the modulator through a 110-GHz radio frequency (RF) probe. A bias-Tee (SHF BT65R-B) was used to apply a reverse voltage of  $-3$  V to the silicon modulator. In Fig. 3b, the measured  $S_{21}$  responses, normalized to the lowest EO response loss, are presented for the carrier wavelength detuning points corresponding to 1, 2, and 3 dB down from the on-resonance transmission maximum power level. The 3-dB EO bandwidths were measured to be 59 GHz, 97 GHz, and  $>110$  GHz at detuning points of  $-1$  dB,  $-2$  dB, and  $-3$  dB, respectively. The EO bandwidth of other devices with the same design were also tested with the experiment results detailed in the Supplementary Information Note 4. The results show a good uniformity, and the bandwidth broadening is available at detuning points of higher optical insertion loss. To analyze the peaking enhancement of the EO response of the PCNC-based modulator, a small-signal model was utilized, similar to that in our previous work<sup>41</sup>. By applying first-order perturbation theory, the small-signal response of the proposed modulator is derived to be

$$S_{21,EO} = \sqrt{\frac{2}{\tau}} \text{Re} \left( \left[ \frac{\frac{1}{\tau} \bar{A} (-iA\delta\omega_r)^*}{i\omega_r - i\omega_0 + i\omega_m + \frac{2}{\tau} + \frac{1}{\tau_r}} + \frac{-i\frac{1}{\tau} A\delta\omega_r (\bar{A})^*}{i\omega_r - i\omega_0 - i\omega_m + \frac{2}{\tau} + \frac{1}{\tau_r}} \right] e^{-i\omega_m t} \right) \quad (5)$$

where  $\delta\omega_r$  is the resonant angular frequency change when a small voltage is applied,  $A$  is the optical field traveling inside the PCNC,  $\tau_r$  is the radiation coupling coefficient between the PCNC and the cladding, and  $\omega_m$ ,  $\omega_0$ , and  $\omega_r$  represent the modulation frequency, input light frequency, and resonance frequency, respectively.

To evaluate the data modulation performance of the silicon modulator, a high-speed data generation experiment was conducted. The experimental setup is shown in Fig. 4a. An electrical data signal with a  $(2^{13}-1)$ -bit pseudorandom binary sequence (PRBS) was applied to the modulator, which was generated by a 256-GSa/s arbitrary waveform generator (AWG, Keysight M8199A) and then amplified by an electrical amplifier (EA, SHF S807C) with a peak-to-peak voltage of  $-2$  V<sub>pp</sub>. A 67-GHz RF cable and a high-speed probe were used to inject the electrical signal into the modulator. A trigger signal was employed for clock recovery to generate eye diagrams of the received data captured by an electrically sampled oscilloscope (OSC, Keysight N1030A). The eye diagrams for the NRZ signals at 50, 70, 90, and 110 Gbps are shown in Fig. 4b–e, with signal-to-noise ratios (SNRs) of 8.23, 6.07, 3.98, and 3.17 dB, respectively. The measured ERs of the open eyes were 1, 0.9, 0.7, and 0.6 dB, respectively. The lower-than-expected dynamic ERs are attributed to the EO response of the proposed PCNC modulator and the electrical signal attenuation<sup>27</sup> through the bias-Tee, RF cable, and high-speed probe, a detailed analysis is illustrated in the Supplementary Information Note 5.

Additionally, a BER measurement under different data rates was performed (see “Methods” for details). Figure 4f shows the BER curves of the 70 Gbps, 90 Gbps, 110 Gbps and 130 Gbps NRZ signals with variations in the receiving power. With properly received optical power, the BERs can drop to well below the hard-decision forward error coding (HD-FEC) threshold ( $3.8 \times 10^{-3}$ ) at a data rate of 110 Gbps and below the soft-decision forward error coding (SD-FEC) threshold ( $2 \times 10^{-2}$ ) at a data rate of up to 130 Gbps. These results demonstrate

**Table 1 | Performance parameters of various state-of-the-art modulators on the silicon platform**

| Year               | Type       | Footprint <sup>a</sup><br>( $\mu\text{m}^2$ ) | Q         | EO BW<br>(GHz) | Dynamic ER (dB) | Tuning efficiency<br>(pm/V) | Data rate                    | Energy<br>(fJ/bit) |
|--------------------|------------|-----------------------------------------------|-----------|----------------|-----------------|-----------------------------|------------------------------|--------------------|
| 2014 <sup>2</sup>  | MRR        | 25                                            | 6600      | 17             | 6               | 250                         | 44 Gbps NRZ                  | 0.9                |
| 2016 <sup>3</sup>  | MRR        | 256                                           | 18,000    | 25             | /               | 16                          | 80 Gbps PAM4<br>45 Gbps PAM8 | 7                  |
| 2020 <sup>25</sup> | MRR        | 100                                           | 3300      | 50             | /               | 30                          | 112 Gbps PAM4                | /                  |
| 2021 <sup>39</sup> | PCNC       | 21                                            | /         | 2.8            | 2.1             | /                           | 5 Gbps NRZ                   | /                  |
| 2021 <sup>40</sup> | PCNC       | 16                                            | 7000      | 38.6           | /               | 10                          | 70 Gbps NRZ                  | /                  |
| 2022 <sup>26</sup> | MRR        | 100                                           | 3000      | 67             | /               | /                           | 220 Gbps PAM4                | /                  |
| 2022 <sup>5</sup>  | MRR        | 336                                           | 4500      | 52–110         | 0.8             | 26.4                        | 240 Gbps PAM8                | /                  |
| 2023 <sup>27</sup> | MOS-MRR    | 900                                           | 3450–4600 | >50            | 3               | 40                          | 64 Gbps NRZ                  | /                  |
| 2023 <sup>23</sup> | Slow-light | 12400 <sup>b</sup>                            | /         | 110            | 3               | /                           | 112 Gbps NRZ                 | /                  |
| 2023 <sup>5</sup>  | MRR        | 1600                                          | 6000      | 0.72           | 68              | $5.6 \times 10^{-3}$        | /                            | /                  |
| 2024 <sup>7</sup>  | MRR        | 576                                           | 3700      | 58.1           | 3.6             | 11/16.3                     | 200 Gbps PAM4                | 6.3                |
| This work          | PCNC       | 10                                            | 2920      | >110           | <1              | 80                          | 110 Gbps NRZ                 | 5.9                |

<sup>a</sup>The device footprint is estimated based on the optical path.

<sup>b</sup>The footprint of the slow-light modulator is limited by the width of the traveling-wave electrode.

the ability of the device to support high-speed data modulation, making it a promising candidate for next-generation high-density photonic integrated circuits.

## Discussion

In this work, we demonstrate a silicon modulator with an EO bandwidth exceeding 110 GHz and an ultracompact footprint of  $10 \mu\text{m}^2$ . This breakthrough is enabled by leveraging the slow-light enhancement effect in a PCNC, coupled with a theoretical framework for designing resonance-based EO modulators with large bandwidths. By carefully optimizing the Q factor of the resonant cavity and improving the tuning efficiency, we achieve an optimal balance between the dynamic ER and EO bandwidth.

While MRR-based modulators on the silicon platform have shown excellent performance, their physical characteristics impose fundamental limitations on further footprint reduction and bandwidth improvement. In contrast, our PCNC-based modulator offers several distinct advantages, including an ultracompact footprint, flexible Q factor control, an ultrasmall mode volume, and seamless integration with optical waveguides. The strong optical confinement in the PCNC also allows a remarkably short PN junction length of only  $10 \mu\text{m}$ , which is instrumental in achieving high-speed modulation. The proposed modulator achieves an EO bandwidth of 110 GHz with a 1.84 dB roll-off and an optical power penalty of 3 dB, which are enabled by the peaking enhancement. High-speed data modulation is successfully achieved for 110 Gbps and 130 Gbps NRZ signal transmissions, with BERs below  $3.8 \times 10^{-3}$  and  $2 \times 10^{-2}$ , respectively, while maintaining a power consumption of 5.9 fJ/bit, calculated as  $CV_{pp}^2/4$ . Nevertheless, the modulator's intrinsic power consumption represents merely a fraction of the total system power (discussed in Supplementary Information Note 6), with the signal driving and recovery processes demanding substantially higher power budgets<sup>53</sup>. Besides, it should be noted that the proposed device represents near-theoretical-limit performance under this design scheme. For practical applications requiring long-distance transmission, high transmission capacity, and low BERs, the device validated in this work can be further optimized in future work.

To contextualize the performance of our device, we compare it with state-of-the-art silicon modulators, as summarized in Table 1. To the best of our knowledge, the demonstrated silicon modulator exhibits a 3-dB bandwidth exceeding 110 GHz with the most compact device footprint reported to date, as illustrated in Fig. 4g. The theoretical dynamic ER of the modulator is estimated to be approximately

2 dB with a  $2 V_{pp}$  voltage swing. However, experimental measurements revealed a dynamic ER of less than 1 dB, primarily because of the EO response of the proposed PCNC modulator and the electrical signal attenuation<sup>27</sup> through the 67-GHz bias-Tee, RF cable, and high-speed probe (discussed in Supplementary Information Note 5). This attenuation resulted in a smaller wavelength shift than anticipated. By employing equipment with reduced electrical attenuation or increasing the voltage swing, the dynamic ER could be significantly improved. Notably, the proposed device was fabricated in a standard multi-project wafer (MPW) process, highlighting its compatibility with large-scale photonic integration without introducing additional fabrication challenges. The scalability underscores the potential of our approach for practical applications in next-generation optical interconnects, high-performance optical computers, and high-bandwidth-density transmitters.

In summary, our work represents a significant advance in silicon photonics, demonstrating a silicon modulator with a bandwidth exceeding 110 GHz and an ultracompact footprint of  $10 \mu\text{m}^2$ . This achievement pushes the boundaries of current silicon modulator technology. Our findings pave the way for the development of next-generation optical communication and computing systems with unprecedented performance and integration density.

## Methods

### Numerical simulation

The band diagrams of the mirror regions and the center cell of the PCNC were simulated by the finite element method. The transmission spectra, Q factor, and mode volume of the PCNC were simulated and optimized using a 3D finite-difference time-domain (FDTD, Lumerical FDTD Solutions) tool with perfectly matched layer boundary conditions.

### Device fabrication

The device was fabricated on a 220-nm SOI platform in a standard CMOS-compatible foundry at AMF in Singapore. The foundry platform provided three etch depths to define the grating couplers, slab waveguides, and strip waveguides using three mask layer processes. The grating couplers and slabs were protected by a photoresist to prevent additional silicon etching during the grating coupler and slab processes. Eight implantations were performed for the high-speed modulators and germanium (Ge) PDs. Additionally, two metal layers and vias were used for electrical connection. The static EO performance of

the fabricated switch was characterized using a tunable laser (Keysight 81960 A), an optical power meter (Keysight N7744A), and a source meter (Keithley 2450). To demonstrate the EO tuning process, a DC probe was used to connect the source meter and electrodes to apply different voltages.

### Setup for high-speed BER measurement

In the high-speed BER test, the experimental equipment was connected as shown in Fig. 4a, with the difference being the specific models of the equipment. A tunable continuous-wave laser (Santec TSL-570) served as the light source. The NRZ signal was applied to the modulator at various baud rates, and this signal was generated by a 256-GSa/s arbitrary waveform generator (AWG, Keysight M8199A) and then amplified by an electrical amplifier (EA, SHF 804 A). A 67-GHz RF cable and a high-speed probe were used to inject the electrical signal into the modulator. A trigger signal was employed for clock recovery to generate eye diagrams of the received data captured by an electrically sampled oscilloscope (OSC, Keysight UXR0704A).

### Data availability

The experimental dataset and its analysis are provided within the paper and its Supplementary Information, and the data are available from the corresponding authors upon reasonable request.

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## Author contributions

H.Z. and Y.K.S. conceived the idea. H.Z. designed and drew the device layout of the EOMs. H.Z. and J.C.L. performed the device characterization and BER experiments. H.Z. and Y.H. performed the numerical simulation. H.W.W. and Y.Z. contributed insightful discussion. H.Z., J.C.L., Y.H., H.W.W., Y.Q.C., Y.Z., X.F.L., Y.H.D., and Y.K.S. participated in writing the manuscript. Y.K.S. supervised this study.

## Competing interests

The authors declare no competing interests.

## Additional information

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